

sercon

User Manual

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sercon manual

Long-form reference for the `pkg/scriptengine` library, the `sercon` CLI, the built-in script API, and the JavaScript runtime surface that `scriptengine` exposes via `goja` and `goja_nodejs`. Examples are runnable — save snippets as `.ts` files and execute with `sercon file.ts` unless otherwise noted.

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1. Overview

`sercon` is a Go program that runs TypeScript files. It's built from two parts:

- `pkg/scriptengine` — a library you embed in your own Go code. You register Go-callable bindings, then call `Run` / `RunFile` to execute a `.ts` script that talks to them.
- `cmd/sercon` — a thin CLI built on the library, useful for ad-hoc test scripts and as a working example of every binding kind.

The runtime is pure Go: `goja` is the JS engine, `esbuild` (used as a Go library) is the TS→JS transpiler, and `goja_nodejs` provides Promises, `setTimeout`, `console`, and CommonJS `require`. No cgo, no Node.

2. Quickstart

Install:

```
go install github.com/codedeviate/sercon/cmd/sercon@latest
```

Run one of the bundled examples:

```
sercon examples/scripts/smoke.ts examples/scripts/async.ts
```

...or run them all via `make demo`. `examples/scripts/` ships a runnable `.ts` (or `.tsx`) file per feature area — `hash.ts`, `strings.ts`, `path-and-time.ts`, `default-export.ts`, `tsx-demo.ts`, and the original `smoke.ts` / `async.ts` / `hang.ts` (`hang.ts` is the timeout demo and intentionally exits non-zero).

Generate a declaration file for editor autocomplete:

```
sercon --emit-dts api.d.ts
```

Embed in your own Go program:

```
eng := scriptengine.New(scriptengine.Options{
    Timeout:    5 * time.Second,
    ScriptRoot: "./scripts",
})
eng.Register("upper", strings.ToUpper)
val, err := eng.RunFile(ctx, "scripts/main.ts")
```

3. Library API — `pkg/scriptengine`

Engine, Options, New

```
type Options struct {
    Timeout      time.Duration // wall-clock limit per Run; 0 disables
    ScriptRoot   string        // base for require/import resolution
    DisableConsole bool          // turn off the goja_nodejs console module
}

func New(opts Options) *Engine
```

`Engine` is the host of all registered bindings. Construct it once, then call `Run` / `RunFile` as many times as needed — each call gets a *fresh* `*goja.Runtime` so script state never leaks between invocations.

`ScriptRoot` defaults to the working directory at `New` time if left empty. Both `Timeout` and `ScriptRoot` can be overridden on a per-Run basis only by constructing a fresh `Engine`; see [Out-of-scope](#) for the per-Run override on the roadmap.

Registering bindings

Function	Use when...
<code>Register(name, value)</code>	The binding is a pure function, struct, or primitive that doesn't need access to the runtime.
<code>RegisterNamespace(name, members map[string]any)</code>	You want to expose <code>name.x</code> , <code>name.y</code> without defining a Go struct.
<code>RegisterConstructor(name, ctor)</code>	The binding is a Go function whose return value should be treated as a class in the emitted <code>.d.ts</code> . (Runtime semantics today are identical to <code>Register</code> ; the <code>d.ts</code> emitter is the only differentiator.)
<code>RegisterFactory(name, factory func(vm, loop) any)</code>	The binding needs the per-Run <code>*goja.Runtime</code> or <code>*eventloop.EventLoop</code> in scope (typical for Promise-returning I/O).
<code>RegisterNamespaceFactory(name, factory)</code>	Same, but the factory returns the full member map.

Example — synchronous function binding:

```
eng.Register("greet", func(name string) string { return "hi " + name })
```

Example — namespace with a sync member:

```
eng.RegisterNamespace("math2", map[string]any{
    "square": func(n float64) float64 { return n * n },
    "pi":     3.14159,
})
```

Example — async (Promise-returning) binding via `PromisifyAsync`:

```

eng.RegisterFactory("httpGet", func(vm *goja.Runtime, loop
*eventloop.EventLoop) any {
    return scriptengine.PromisifyAsync(vm, loop,
        func(ctx context.Context, call goja.FunctionCall) (map[string]any,
error) {
            url := call.Argument(0).String()
            resp, err := http.Get(url)
            if err != nil {
                return nil, err
            }
            defer resp.Body.Close()
            body, _ := io.ReadAll(resp.Body)
            return map[string]any{"status": resp.StatusCode, "body":
string(body)}, nil
        })
})

```

Run, RunFile

```

func (e *Engine) Run(ctx context.Context, name, source string, opts
...RunOption) (goja.Value, error)
func (e *Engine) RunFile(ctx context.Context, path string, opts
...RunOption) (goja.Value, error)

```

`Run` executes `source` as an entry-script TS. `name` is used in stack traces. The returned value is currently always `undefined` — top-level expression capture is on the backlog.

Both methods respect `Options.Timeout` and `ctx` cancellation, whichever fires first; the resulting error is either `ErrScriptTimeout`, `ctx.Err()`, or the underlying JS exception.

Per-Run options

```

type RunOption func(*runConfig)

func WithScriptRoot(dir string) RunOption

```

Reuse a single Engine across many scripts that each live in their own directory by passing `WithScriptRoot(dir)` to `Run` / `RunFile`. The override applies only to this call; `Options.ScriptRoot` is used for every other `Run`.

```

_, _ = eng.Run(ctx, "main.ts", source,
scriptengine.WithScriptRoot("/path/to/run42"))

```

Reset

```
func (e *Engine) Reset()
```

Clears every registered binding. Useful when a long-lived Engine is reused across unrelated batches of scripts that each want a clean global namespace. **Not safe to call concurrently with Run / RunFile.**

WriteTypes

```
func (e *Engine) WriteTypes(w io.Writer) error
```

Emits a `.d.ts` describing the registered surface. See section [13. Type generation](#) for what the mapping looks like.

SetDocs and SetMemberDocs

```
func (e *Engine) SetDocs(path, doc string)
func (e *Engine) SetMemberDocs(namespace string, docs map[string]string)
```

Attach JSDoc strings to registered bindings so the emitted `.d.ts` grows readable editor hover. `SetDocs` takes a dotted path — either a bare top-level name (`"greet"`, `"http"`) or `"namespace.member"` for namespace members (`"http.get"`, `"exec.shell"`); `SetMemberDocs` is the convenience for documenting many members of a namespace at once (the namespace itself can still be documented separately via `SetDocs`).

Multi-line docs are written as `\n`-separated strings; the emitter expands them into a standard `* -`prefixed JSDoc block. Single-line docs collapse to `/** ... */`. Calling `SetDocs` with an empty string removes any previous doc for that path. Bindings without a doc entry get no JSDoc block at all — the emitter doesn't insert empty `/** */` placeholders.

`SetDocs` may be called before or after the matching `Register...` call; docs are looked up by name at emit time. Like the registration methods, `SetDocs` must not race with a `Run` / `WriteTypes` / `Reset` on the same engine.

```

eng.Register("greet", func(name string) string { return "hi " + name })
eng.SetDocs("greet", "Greet someone by name.")

eng.RegisterNamespace("math2", map[string]any{
    "pi":      3.14159,
    "squared": func(n float64) float64 { return n * n },
})
eng.SetDocs("math2", "Tiny math helpers.")
eng.SetMemberDocs("math2", map[string]string{
    "pi":      "Circumference / diameter ratio.",
    "squared": "Multiply a number by itself.",
})

```

PromisifyAsync[T] and AsyncBinding

```

type AsyncBinding struct {
    Func      func(goja.FunctionCall) goja.Value
    TSReturnType string
}

func PromisifyAsync[T any](vm *goja.Runtime, loop *eventloop.EventLoop,
    work func(ctx context.Context, call goja.FunctionCall) (T, error),
) AsyncBinding

```

`PromisifyAsync` turns blocking Go work into a JS Promise. It launches the work in a goroutine, parks a `SetTimeout` sentinel so `loop.Run` doesn't return early, and schedules the resolve/reject back onto the event loop. **Required** for any Promise-returning binding — `RunOnLoop` alone is not counted as a live job by the event loop.

`PromisifyAsync` returns an `AsyncBinding` carrier rather than the bare callback. The engine unwraps it to `.Func` at `vm.Set` time so goja's special-case detection of `func(goja.FunctionCall) goja.Value` host-callbacks still fires; the `.d.ts` emitter reads `.TSReturnType` to emit `Promise<T>` (mapped from the generic `T` via reflect at construction time) instead of the previous `unknown`. Hosts assigning the result into a `map[string]any` for `RegisterNamespace` / `RegisterNamespaceFactory` get the unwrap recursively too — no manual work required.

Version

```

import "github.com/codedeviate/sercon/pkg/scriptengine"

fmt.Println(scriptengine.Version) // "0.1.0"

```

Bumped in lockstep with the git tag.

4. CLI — sercon

```
sercon [flags] <script.ts> [script.ts ...]
sercon [flags] -                # read entry script from stdin
sercon --examples | --help | --version
```

Flag	Purpose
<code>-timeout DURATION</code>	Wall-clock limit per script (default <code>10s</code> ; <code>0</code> disables).
<code>-root DIR</code>	Override the script root for <code>require</code> / <code>import</code> resolution.
<code>-emit-dts PATH</code>	Write the example bindings' <code>.d.ts</code> to <code>PATH</code> and exit.
<code>-v</code>	Verbose: trace the rewritten entry-script JS and each module resolution to stderr; also print duration on failure.
<code>-h</code> , <code>--help</code>	In-depth colorized help.
<code>--examples</code>	In-depth colorized walkthrough of every feature.
<code>--version</code>	Print the engine version (plus goja/esbuild build-info versions).

Each positional argument is either a path to a `.ts` / `.tsx` file or `-` to read an entry script from standard input:

```
echo 'api.log(1 + 2);' | sercon -
sercon prelude.ts -          # prelude then stdin
```

Exit codes are distinct per failure type so shells can react sensibly. When several scripts run, the highest applicable code wins:

Code	Meaning
<code>0</code>	every script passed.
<code>1</code>	CLI usage error (unknown flag, missing script argument, ...).
<code>2</code>	at least one script failed to transpile — never ran.
<code>3</code>	at least one script timed out (<code>-timeout</code>) or was context-cancelled.
<code>4</code>	at least one script ran and threw a JS exception.

`-v` writes lines prefixed with `[sercon]` to stderr. The traces include the full rewritten entry-script JS (the form goja actually runs, after the ESM→CJS rewrite + async IIFE wrapper) and every module-resolution event, so debugging an unexpected resolve target or a mis-rewritten import is straightforward.

The CLI registers a single `api` namespace; see the next section.

5. Built-in script `api`

The CLI exposes the following globals to every script:

```

declare const api: {
  log(...args: unknown[]): void;

  assert: {
    equal(actual: unknown, expected: unknown, msg?: string): void;
    ok(cond: unknown, msg?: string): void;
  };

  http: {
    get(url: string): Promise<{ status: number; body: string }>;
    post(url: string, body?: string): Promise<{ status: number; body:
string }>;
  };

  time: {
    nowMs(): number;
    sleep(ms: number): Promise<void>;
    format(unixMs: number, fmt: string, tz?: string): string;
  };

  env: {
    get(name: string): string | undefined;
  };

  str: {
    trim(s: string, mask?: string): string;
    ltrim(s: string, mask?: string): string;
    rtrim(s: string, mask?: string): string;
    reverse(s: string): string;
    stripHtml(s: string): string;
    nl2br(s: string, xhtml?: boolean): string;
    br2nl(s: string): string;
    base64Encode(s: string): string;
    base64Decode(s: string): string;
    urlEncode(s: string): string;
    urlDecode(s: string): string;
    htmlEntityDecode(s: string): string;
    pad(s: string, len: number, padChar?: string, side?: "left" | "right" |
"both"): string;
    lpad(s: string, len: number, padChar?: string): string;
    rpad(s: string, len: number, padChar?: string): string;
    sprintf(fmt: string, ...args: unknown[]): string;
    printf(fmt: string, ...args: unknown[]): void;
    normalizeNewlines(s: string, style: "lf" | "crlf" | "cr"): string;
  };

  path: {
    dirname(p: string): string;
  };
};

```

```

    basename(p: string, suffix?: string): string;
};

hash: {
    md5(data: string): string;
    sha1(data: string): string;
    sha256(data: string): string;
    sha384(data: string): string;
    sha512(data: string): string;
    sha3_256(data: string): string;
    sha3_512(data: string): string;
    blake3(data: string): string;
    crc32(data: string): string;
};

compression: {
    algos(): string[];
    compress(
        algo: "gzip" | "deflate" | "zlib" | "bzip2" | "zstd" | "brotli" |
        "lz4" | "xz" | "snappy",
        data: string | ArrayBuffer | Uint8Array,
    ): Promise<Uint8Array>;
    decompress(
        algo: "gzip" | "deflate" | "zlib" | "bzip2" | "zstd" | "brotli" |
        "lz4" | "xz" | "snappy",
        data: string | ArrayBuffer | Uint8Array,
    ): Promise<Uint8Array>;
};

barcode: {
    formats(): string[];
    decodableFormats(): string[];
    encode(
        format: "qr" | "datamatrix" | "aztec" | "pdf417"
            | "code128" | "code39" | "codabar"
            | "ean13" | "ean8" | "upca",
        data: string,
        opts?: { width?: number; height?: number },
    ): Promise<Uint8Array>;
    decode(
        data: Uint8Array | ArrayBuffer | string,          // PNG / JPEG / WebP
        image bytes
        format?: "qr" | "datamatrix" | "aztec"
            | "code128" | "code39" | "code93" | "codabar"
            | "ean13" | "ean8" | "upca" | "upce" | "itf",
    ): Promise<{ format: string; text: string }>;
};

text: {

```

```

    detect(data: string | ArrayBuffer | Uint8Array): Promise<{
      charset: string;
      confidence: number;
      language?: string;
      candidates: Array<{ charset: string; confidence: number; language?:
string }>;
    }>;
    decode(data: string | ArrayBuffer | Uint8Array, charset: string):
Promise<string>;
    encode(text: string, charset: string): Promise<Uint8Array>;
  };

  checkdigit: {
    algos(): string[];
    validate(
      algo: "luhn" | "isbn10" | "isbn13" | "ean13" | "ean8" | "upca",
      input: string,
    ): boolean;
    compute(
      algo: "luhn" | "isbn10" | "isbn13" | "ean13" | "ean8" | "upca",
      partial: string,
    ): string;
    inspect(
      algo: "luhn" | "isbn10" | "isbn13" | "ean13" | "ean8" | "upca",
      input: string,
    ): {
      algo: string;
      input: string;
      valid: boolean;
      given: string;
      computed: string;
    };
  };

  archive: {
    create(
      destPath: string,
      sources: Array<string | { path: string; name?: string }>,
    ): Promise<{ path: string; format: string; entries: string[]; bytes?:
number }>;
    extract(
      archivePath: string,
      destDir: string,
      opts?: { overwrite?: boolean },
    ): Promise<{ path: string; format: string; dest: string; entries:
string[] }>;
  };

  diff: {

```

```

compare(
  a: string | ArrayBuffer | Uint8Array,
  b: string | ArrayBuffer | Uint8Array,
  opts?: { context?: number; fromFile?: string; toFile?: string },
): Promise<{
  identical: boolean;
  binary: boolean;
  added: number;
  removed: number;
  diff: string;
  format: "unified";
}>;
};

jq: {
  query(data: unknown, filter: string): Promise<unknown>;
  queryAll(data: unknown, filter: string): Promise<unknown[]>;
};

net: {
  tcp(target: string, opts?: { timeout?: number; port?: string }):
Promise<{
    host: string; port: number; ip: string; latencyMs: number;
  }>;
  dns(host: string, opts?: { types?: ("a" | "aaaa" | "mx" | "txt" |
"cname" | "ns")[] }): Promise<{
    a?: string[]; aaaa?: string[];
    mx?: { preference: number; host: string }[];
    txt?: string[]; cname?: string; ns?: string[];
  }>;
  tls(target: string, opts?: { timeout?: number }): Promise<{
    cn: string; issuer: string;
    notBefore: string; notAfter: string;
    daysRemaining: number;
    dnsNames: string[];
    serialNumber: string;
    fingerprintSha256: string;
  }>;
  ntp(host: string, opts?: { timeout?: number; port?: number }):
Promise<{
    serverTime: string;
    offsetMs: number;
    rttMs: number;
    stratum: number;
    referenceTime: string;
    rootDelayMs: number;
    rootDispersionMs: number;
  }>;
  whois(domain: string, opts?: { timeout?: number }): Promise<{

```

```

    raw: string;
    domain?: {
      name: string; punycode?: string;
      whoisServer?: string;
      nameServers?: string[]; status?: string[];
      dnssec?: boolean;
      createDate?: string; updatedAt?: string; expirationDate?:
string;
    };
    registrar?: { name?: string };
  }>;
};

exec: {
  shell(
    cmd: string | string[],
    opts?: {
      cwd?: string;
      env?: Record<string, string>;
      timeout?: number; // ms; default 30000
      stdin?: string;
    },
  ): Promise<{
    stdout: string;
    stderr: string;
    exitCode: number;
    success: boolean;
    durationMs: number;
  }>;
  http(
    method: string,
    url: string,
    opts?: {
      headers?: Record<string, string>;
      body?: string;
      timeout?: number; // ms; default 30000
      follow?: boolean; // -L
      insecure?: boolean; // -k
      backend?: "auto" | "recon" | "curl"; // default "auto"
    },
  ): Promise<{
    status: number;
    headers: Record<string, string>;
    body: string;
    durationMs: number;
    backend: "recon" | "curl";
  }>;
};

```

```

git: {
  branch(opts?: { cwd?: string }): Promise<{
    current: string;          // branch name; empty on detached HEAD
    detached: boolean;
    all: string[];           // local branches
  }>;
  isClean(opts?: { cwd?: string }): Promise<boolean>;
  revParse(rev: string, opts?: { cwd?: string }): Promise<string>;
  status(opts?: { cwd?: string }): Promise<Array<{
    path: string;
    indexStatus: string;      // single-char porcelain code
    workingStatus: string;
  }>>;
  add(paths: string | string[], opts?: { cwd?: string }): Promise<{
paths: string[] }>;
    commit(message: string, opts?: { cwd?: string; allowEmpty?: boolean }):
Promise<{ sha: string }>;
    log(opts?: { cwd?: string; limit?: number; revRange?: string }):
Promise<Array<{
      sha: string;
      shortSha: string;
      author: string;
      email: string;
      timestamp: number;      // unix seconds
      subject: string;
    }>>;
    diffStat(opts?: { cwd?: string; revRange?: string }): Promise<{
      files: number;
      insertions: number;
      deletions: number;
    }>;
    runText(args: string[], opts?: { cwd?: string }): Promise<{
      stdout: string;
      stderr: string;
      exitCode: number;       // surfaced as data – does NOT throw
    }>;
  };

  preg: {
    match(pattern: string, subject: string): {
      match: string;
      groups: string[];      // empty string for an optional group that
didn't match
      index: number;
    } | null;
    matchAll(pattern: string, subject: string): Array<{
      match: string;
      groups: string[];
      index: number;
    }>;
  };

```

```

    }>;
    replace(pattern: string, replacement: string, subject: string): string;
};

jwt: {
    sign(
        claims: Record<string, unknown>,
        secret: string, // raw bytes for HS*; PEM-
encoded private key for RS*/PS*/ES*/EdDSA
        opts?: {
            algorithm?:
                | "HS256" | "HS384" | "HS512" // HMAC
                | "RS256" | "RS384" | "RS512" // RSA-PKCS1v1.5
                | "PS256" | "PS384" | "PS512" // RSA-PSS
                | "ES256" | "ES384" | "ES512" // ECDSA
                | "EdDSA"; // Ed25519
            // default "HS256"
        },
    ): string;
    view(token: string): {
        header: Record<string, unknown>;
        payload: Record<string, unknown>;
        signature: string; // raw base64url, no
padding
    };
    validate(
        token: string,
        secret: string, // raw bytes for HS*; PEM-
encoded public key (or certificate) for asymmetric
        opts?: {
            algorithm?:
                | "HS256" | "HS384" | "HS512"
                | "RS256" | "RS384" | "RS512"
                | "PS256" | "PS384" | "PS512"
                | "ES256" | "ES384" | "ES512"
                | "EdDSA";
            audience?: string;
            issuer?: string;
        },
    ): { valid: true; claims: Record<string, unknown> }
        | { valid: false; reason: string };
};

encrypt: {
    keygen(): { publicKey: string; privateKey: string };
    encrypt(
        data: string | Uint8Array | ArrayBuffer,
        recipients: string | string[], // age1... public keys
        opts?: { armored?: boolean }, // default false → binary age
format
    );
};

```



```

    ): Uint8Array;
    decrypt(
        ciphertext: string | Uint8Array | ArrayBuffer,
        identities: string | string[],           // AGE-SECRET-KEY-1...
private keys
    ): Uint8Array;                             // auto-detects binary vs
armored
    rekey(
        ciphertext: string | Uint8Array | ArrayBuffer,
        oldIdentities: string | string[],
        newRecipients: string | string[],
        opts?: { armored?: boolean },           // default: preserve input
format
    ): Uint8Array;
    detectBackend(input: string):
        | { backend: "age";      kind: "public" | "private" }
        | { backend: "pgp";      kind: "public" | "private" }
        | { backend: "unknown" };
    };

    gh: {
        authStatus(): Promise<{
            authenticated: boolean;
            user: string;           // login or empty
            raw: string;           // login on success, stderr / probe note
on failure
        }>;
        prList(opts?: {
            cwd?: string;
            state?: "open" | "closed" | "merged" | "all";
            limit?: number;         // default 30
            author?: string;
        }): Promise<Array<{
            number: number;
            title: string;
            state: string;
            author: string;         // login (flattened from gh's wrapper)
            headRefName: string;
            baseRefName: string;
            url: string;
            createdAt: string;
            updatedAt: string;
        }>>;
        repoView(
            repo?: string,          // "owner/name"; omit to use cwd's repo
            opts?: { cwd?: string },
        ): Promise<{
            name: string;
            owner: string;         // login (flattened)

```

```

    description: string;
    url: string;
    defaultBranch: string;    // flattened from defaultBranchRef.name
    visibility: string;
  }>;
};

email: {
  spf(domain: string): Promise<
    | { present: false }
    | {
      present: true;
      record: string;
      mechanisms: string[];
      allPolicy: "pass" | "fail" | "softfail" | "neutral" | "";
    }
  >;
  dmarc(domain: string): Promise<
    | { present: false }
    | {
      present: true;
      record: string;
      tags: Record<string, string>;
      policy?: string;
      subdomain?: string;
      percent?: string;
      rua?: string;
      ruf?: string;
    }
  >;
  mtaSts(domain: string): Promise<
    | { present: false }
    | {
      present: true;
      record: string;
      txt: { v: string; id: string };
      policy?: {
        version?: string;
        mode?: "enforce" | "testing" | "none" | string;
        mx?: string[];
        maxAge?: number | string;
      };
      policyError?: string;
    }
  >;
  tlsRpt(domain: string): Promise<
    | { present: false }
    | {
      present: true;

```

```

        record: string;
        tags: Record<string, string>;
        rua?: string;
    }
>;
bimi(domain: string, opts?: { selector?: string }): Promise<
    | { present: false; selector: string }
    | {
        present: true;
        selector: string;
        record: string;
        tags: Record<string, string>;
        l?: string;
        a?: string;
    }
>;
all(domain: string): Promise<{
    domain: string;
    spf:    unknown;
    dmarc:  unknown;
    mtaSts: unknown;
    tlsRpt: unknown;
    bimi:   unknown;
}>;
};
};

```

Examples:

```

api.log("hello", 1 + 2);
api.assert.equal(api.time.nowMs() > 0, true);

const r = await api.http.get("https://example.com");
api.assert.ok(r.status === 200, `expected 200, got ${r.status}`);

await api.time.sleep(50);
const home = api.env.get("HOME") ?? "(none)";

api.log(api.hash.sha256("abc"));
// → ba7816bf8f01cfea414140de5dae2223b00361a396177a9cb410ff61f20015ad

```

HTTP bindings use `net/http` with a 5-second default per-request timeout and surface real `Promise<...>` values through the event loop. They are *not* mockable from JS — they go to the real network.

Compression bindings (`api.compression.*`) cover nine pure-Go algorithms behind a uniform interface. Inputs are either strings (interpreted as UTF-8 byte sequences) or any `ArrayBuffer` / `Uint8Array`; outputs are `Uint8Array` — goja's representation of a Go `[]byte` return is a

`Uint8Array`, so scripts can read `.length` and iterate directly without an `ArrayBuffer` view. Every algorithm round-trips:

Algorithm	Library
<code>gzip / deflate / zlib</code>	<code>stdlib compress/*</code>
<code>bzip2</code>	<code>stdlib compress/bzip2</code> for read; github.com/dsnet/compress/bzip2 for write
<code>zstd</code>	github.com/klauspost/compress/zstd
<code>brotli</code>	github.com/andybalholm/brotli
<code>lz4</code>	github.com/pierrec/lz4/v4
<code>xz</code>	github.com/ulikunitz/xz
<code>snappy</code>	github.com/golang/snappy

`api.compression.algos()` returns the supported names so scripts can iterate without hard-coding the list. Unknown algorithm names throw a clear error rather than silently returning empty bytes.

Barcode encoders (`api.barcode.*`) cover ten symbologies through one `encode(format, data, opts?)` entry point, all backed by the pure-Go github.com/boombuler/barcode toolkit. Output is a PNG payload as `Uint8Array`; size defaults to `256x256` for 2D codes (QR / DataMatrix / Aztec) and `400x120` for linear ones, overridable via `opts.width / opts.height`.

- 2D: `qr` (medium error correction, auto mode), `datamatrix`, `aztec` (33% ECC, auto layers), `pdf417` (security level 5).
- Linear: `code128`, `code39` (Mod-43 checksum, ASCII-only), `codabar`.
- Retail: `ean13`, `ean8`, `upca` — content must be numeric and the right length; `boombuler/ean.Encode` dispatches by length so all three variants share the same encoder.

`api.barcode.formats()` returns the supported encode names;

`api.barcode.decodableFormats()` returns the (slightly different) decode set.

Decoding (`api.barcode.decode(data, format?)`) accepts PNG / JPEG / WebP image bytes and reaches into github.com/makiuchi-d/gozxing. Returns `{ format, text }` — `format` is the snake-case label the encoder accepts, so round-trips line up. WebP support comes via a blank-import of `golang.org/x/image/webp` that registers the decoder with `stdlib's image.Decode`.

- **With a `format` hint:** only that reader runs. Fast, and a mismatch surfaces a clear error rather than a silent miss.

- **Without a hint:** every reader in a priority list runs (2D formats first; the more-constrained 1D readers — UPC/EAN — before the permissive ones — Code 128 / Codabar). First success wins; if no reader matches, the call throws with "no barcode recognised".

Decode quirks worth knowing — these are properties of the underlying libraries, not sercon bugs:

- **pdf417 isn't decodable.** gozxing v0.1.1 has no PDF417 reader; pass it as a hint and the call throws "unsupported format". The encoder still emits PDF417 just fine.
- **EAN / UPC need a quiet zone.** Per ISO/IEC 15420 + 15418, retail symbologies require white margin on both sides of the bars. boombuler's encoder doesn't add one, so a `barcode.encode("ean13", x) → barcode.decode(...)` round-trip fails on the decode step. Real-world EAN scanners need that margin too. Wrap the encoded PNG in a wider white canvas (or use a different encoder) when round-trips matter.
- **Code 39 decoded text ends with the Mod-43 checksum char.** boombuler's encoder appends it; gozxing returns it.
- **Codabar wraps payload in start/stop chars** like `A...A` on encode; gozxing strips them on decode.

Charset bindings (`api.text.*`) cover detection plus byte-side round-tripping:

- **detect(data)** — runs [saintfish/chardet](#) over the input and returns the top guess (charset, confidence on a 0–100 scale, optional language hint) plus the full candidate list. Input is bytes (string, `ArrayBuffer`, or `Uint8Array`).
- **encode(text, charset)** — converts a UTF-8 string to bytes in the target encoding. Characters with no representation in the target encoding cause the encoder to error out rather than silently lose them; pre-process the input yourself if you want lossy behaviour.
- **decode(data, charset)** — the inverse: bytes-in-charset → UTF-8 string.

Charset names follow the WHATWG Encoding Living Standard aliases that `golang.org/x/text/encoding/htmlindex.Get` accepts — UTF-8, ISO-8859-1, Windows-1252, Shift_JIS, GBK, GB18030, Big5, EUC-JP, EUC-KR, etc., plus all documented aliases. Unknown names throw a clear error.

Check-digit bindings (`api.checkdigit.*`) verify and compute the trailing check digit of common numeric codes. All three members are synchronous — the algorithms are small bits of modular arithmetic, no I/O or library dependency. Supported algorithms (returned by `algorithms()`):

Algorithm	Input length	Notes
luhn	any ≥ 2	Credit cards, IMEI, social-security-style IDs. Mod-10 with right-to-left doubling.
isbn10	10	Last position may be X (= 10). Mod-11 with weighted sum.
isbn13	13	Alias of ean13 — same algorithm; the 978 / 979 prefix is the only thing that distinguishes ISBN-13 syntactically.

Algorithm	Input length	Notes
ean13	13	Mod-10, weights 1, 3, 1, 3, ... from position 0.
ean8	8	Mod-10, weights 3, 1, 3, 1, ...
upca	12	Mod-10, weights 3, 1, 3, 1, ...

`validate(algo, input)` returns a plain boolean — non-digit characters, wrong length, and unknown algorithm names all surface as `false` so scripts can run quick presence checks without try/catch. `compute(algo, partial)` takes the input *without* its check digit and returns just that digit (or "X" for ISBN-10 position 10). `inspect(algo, input)` returns the union of both views — useful when you want to display the diagnosis to a human.

Archive bindings (`api.archive.*`) handle the three formats `stdlib` provides without external deps: `.zip`, `.tar`, and `.tar.gz` (also `.tgz`). Format is inferred from the destination's extension on `create` and from the source path on `extract`.

- **`create(destPath, sources)`** — `sources` is an array of either bare paths (the basename is used as the in-archive name) or `{ path, name }` objects (override the in-archive name). Directory sources are recursed; the directory's basename becomes the archive subdir and the rest of the tree is added relative to it. Returns the count + list of entries and the output file size.
- **`extract(archivePath, destDir, opts?)`** — unpacks into `destDir`. `opts.overwrite` (default `false`) controls whether existing files at the destination are clobbered; on `false` a collision errors out (`os.ErrExist` -shaped). Every entry name is run through a zip-slip / tar-slip guard that rejects absolute paths, `..` segments, and anything that would resolve outside `destDir`.

Diff bindings (`api.diff.compare(a, b, opts?)`) produce a unified diff between two text inputs via github.com/pmezard/go-difflib. Inputs are strings (UTF-8) or any byte sequence (`ArrayBuffer` / `Uint8Array`). The result reports:

- `identical` — byte-equal inputs short-circuit; `diff` is empty.
- `binary` — either input has a NUL byte in its first 8 KB. A unified diff of binary content isn't useful so `diff` is empty.
- `added` / `removed` — body-only `+` / `-` line counts (file headers are excluded so the numbers match `git diff --shortstat`).
- `diff` — the unified diff text, or empty when `identical` or `binary` is `true`.
- `format` — always "unified" today; reserved for future alternatives (`"side-by-side"`, `"summary"`, ...).

`opts.context` (default 3) is the context-line count; `opts.fromFile` / `opts.toFile` (default `"a"` / `"b"`) set the labels in the diff headers.

JSON-query bindings (`api.jq.*`) run jq filters over JS data via github.com/itchyny/gojq, a pure-Go re-implementation. The filter syntax is the same as command-line jq.

- **query(data, filter)** — runs the filter and returns the first emitted value (or `null` when the filter emits nothing).
- **queryAll(data, filter)** — drains the result iterator and returns every emitted value as an array.

Data passes through goja's `Export` as `map[string]any` / `[]any` trees, which gojq's runtime accepts directly. There's a small normalisation pass on the way in that converts all sized integer types (`int64`, `int32`, etc.) to plain `int` and `float32` to `float64` — gojq's arithmetic dispatch only knows the two canonical numeric kinds, and goja exports every JS-side integer as `int64`. Without normalisation a query as simple as `.users[].age` would panic in gojq.

Parse errors and runtime errors (type mismatches, division by zero, ...) both surface as JS exceptions. Use jq's optional access operator `?` (e.g. `.does.not.exist?`) to suppress missing-path errors and get `null` back instead.

Subprocess bindings (`api.exec.*`) wrap Go's `os/exec`. `shell` is the only entry today; `http` (recon-with-curl-fallback) and `git` / `gh` wrappers ride on top of it in later 0.4.x cuts.

- **shell(cmd, opts?)** — runs a subprocess and waits for it to exit. `cmd` is either a **string** (passed verbatim to `/bin/sh -c` on Unix or `cmd /C` on Windows so pipes, redirects, and globs work as typed) or a **string[]** (treated as `argv`, no shell — use this when arguments could be re-interpreted by the shell). Returns `{ stdout, stderr, exitCode, success, durationMs }`.

Options: `cwd` sets the working directory; `env` is merged on top of the parent process environment (not a replacement); `timeout` is in milliseconds and defaults to **30 000 ms**; `stdin` is piped to the child's standard input.

Process-start failures (host binary not on PATH, permission denied) throw. Context deadline (`timeout`) and engine cancellation throw too — the subprocess never got to choose its own exit code, so it isn't reasonable to surface a fake one. Non-zero exits do **not** throw: `success: false` + the real `exitCode` is what callers want for the usual "ran the linter, expected to be told it failed" flow.

- **http(method, url, opts?)** — curl-compatible HTTP client routed through `recon` (preferred) with `curl` as a fallback. Returns `{ status, headers, body, durationMs, backend }`. The choice between the two binaries is normally invisible to scripts — both implement the same curl-style flag set we use here (`-X`, `-H`, `-D`, `--data-binary`, `-L`, `-k`, `-s`). Pick a specific backend with `opts.backend: "recon" | "curl"` when you need to.

Implementation notes: response headers are dumped to a temp file via `-D <path>` and parsed back, rather than relying on `-i`'s body-stream format (recon's `-i` is verbose-debug style with `<` prefixes, incompatible with curl's wire-format `-i`). Request bodies are materialised to a temp file and passed via `--data-binary @<path>` so CR / LF are preserved regardless of backend. Header names in the returned map are lower-cased.

On redirect chains (`opts.follow: true`), the *last* response wins — the intermediate 3xx blocks are skipped.

HTTP 4xx / 5xx do **not** throw — they're a normal HTTP outcome and callers branch on `status`. Process-start failures, transport errors (DNS, connection refused, TLS handshake), and context deadline / cancellation throw. The error message includes the backend's `stderr` when present.

Git bindings (`api.git.*`) shell out to the host `git` binary; no pure-Go alternative (`go-git` is heavier and would drift from the user's installed `git`). Every binding accepts an `opts.cwd` so a single engine can work across multiple checkouts.

- **`branch(opts?)`** — Reports the current branch (empty string when HEAD is detached) plus every local branch from `for-each-ref refs/heads`. `detached` is true exactly when `symbolic-ref --short HEAD` exited non-zero, which is `git`'s own signal for the state.
- **`isClean(opts?)`** — Convenience boolean over `git status --porcelain`. True when the porcelain output is empty.
- **`revParse(rev, opts?)`** — Returns the 40-char SHA for `rev`. Invalid refs throw with `git`'s `stderr` message included.
- **`status(opts?)`** — Parses `--porcelain v1` output into `{ path, indexStatus, workingStatus }`. The two single-char status codes are the raw porcelain codes (M , A , D , R , ? , ...) so scripts can dispatch on them without re-parsing strings.
- **`add(paths, opts?)`** — Stages one path (string) or a list (string[]). `--` is inserted between `add` and the paths so values that look like flags work too. Returns the resolved paths array.
- **`commit(message, opts?)`** — Creates a new commit; the post-commit HEAD SHA is returned. `allowEmpty:true` toggles `--allow-empty`. An empty message throws before spawning.
- **`log(opts?)`** — Returns `limit` (default 50) most-recent commits in `revRange` (default HEAD). Format string `%H%x09%h%x09%an%x09%ae%x09%at%x09%s` tab-separates SHA, short SHA, author name, email, unix timestamp, and subject. Newlines and tabs in subjects are already collapsed by `git` so the parse stays one-line-per-commit.
- **`diffStat(opts?)`** — Aggregates `git diff --shortstat` into `{ files, insertions, deletions }`. Default range is `HEAD~1..HEAD`. A pure-add or pure-delete diff returns zero for the missing side instead of throwing.
- **`runText(args, opts?)`** — Escape hatch. Surfaces `{ stdout, stderr, exitCode }` for any `git <args>` invocation. Non-zero exits do **not** throw — that's the whole point of having a generic wrapper; callers branch on `exitCode`. Spawn failures and context cancellation still throw.

GitHub-CLI bindings (`api.gh.*`) wrap the `gh` binary. They respect whatever authentication state `gh auth` is already in; we don't try to swap accounts or manage tokens. Every call uses `gh --json` so the result is structured rather than parsed out of human-readable text.

- **authStatus()** — Probe whether gh is installed *and* authenticated. Under the hood this runs `gh api user --jq .login` (machine-friendly: just the login on success, a clear error otherwise) rather than `gh auth status` (multi-line human report). Missing-gh and unauthenticated-session both resolve with `{ authenticated: false, ...}` rather than throwing — the whole point of a status probe is that the script branches on it. Only context cancellation throws.
- **prList(opts?)** — Lists pull requests on the repo identified by `opts.cwd` (or the engine's working directory). Defaults: open state, limit 30. `gh`'s `author: { login, ... }` wrapper is flattened to just the login string so scripts can compare directly.
- **repoView(repo?, opts?)** — Returns metadata about a repo. With no argument it asks `gh` about the `cwd`'s repo (works from inside a checkout); pass `"owner/name"` to look up any repo `gh` has access to. Two convenience flattenings are applied: `owner` is a login string (not `{ login, ... }`), and `defaultBranch` is the branch name (not `defaultBranchRef.name`). Empty repos resolve with `defaultBranch: ""` rather than `undefined`.

Regex bindings (`api.preg.*`) accept PHP-style `/pattern/flags` delimited input but run on Go's `stdlib regexp` (RE2). The "preg" naming is a deliberate homage; the semantics are RE2's. That means **no backreferences inside patterns, no lookahead / lookbehind, and no possessive quantifiers** — scripts that need those should reach for goja's native `RegExp` (ECMAScript-flavoured) or wait for a dedicated PCRE-compatible binding (`regexp2` is on the backlog).

Supported flags: `i` (case-insensitive), `m` (multiline — `^` and `$` match line boundaries), `s` (dotall — `.` matches newline). The PHP flags `u` (Unicode), `U` (default-ungreedy), and `x` (extended whitespace-ignoring) all surface a clear error rather than silently dropping; unknown flags do too. RE2 is UTF-8 by default, so `u` is unnecessary in any case — the error message says so.

- **match(pattern, subject)** — Returns `{ match, groups, index }` for the first hit, or `null` when nothing matches. `groups` is every numbered submatch after group 0, in order; an optional group that didn't match surfaces as an empty string (not `undefined`) so the result type stays stable.
- **matchAll(pattern, subject)** — Same shape, drained: returns an array of match objects, one per hit.
- **replace(pattern, replacement, subject)** — Substitutes via Go's `$1 / ${1}` backreference syntax. PHP's `\1` form is **not** translated — that would conflict with `\\` escapes that are already legitimate in the replacement. Use `$1 / ${1}` as in Go.

JWT bindings (`api.jwt.*`) wrap [golang-jwt/jwt/v5](https://github.com/golang-jwt/jwt/v5). The full RFC 7518 algorithm matrix is supported: HMAC (`HS256 / HS384 / HS512`), RSA-PKCS1v1.5 (`RS256 / RS384 / RS512`), RSA-PSS (`PS256 / PS384 / PS512`), ECDSA (`ES256 / ES384 / ES512`), and Ed25519 (EdDSA). Unsupported algorithm names — including the special `"none"` value, garbage like `"HS999"`, and protocol names that aren't JWT signing algos — throw at `sign / validate` time with a named-algorithm error rather than silently falling through.

Key shape. The `secret` parameter is overloaded by the algorithm: HMAC algorithms use the byte string directly; asymmetric algorithms expect a PEM-encoded key (private for `sign`, public — or a certificate the public key can be pulled from — for `validate`). PEM detection is the literal `-----BEGIN` prefix. Both directions of the cross-check are enforced:

- **PEM secret + HMAC algorithm** throws with a named-algorithm hint ("looks like a PEM-encoded key — set `opts.algorithm` to RS256 / ES256 / EdDSA / etc."). Without this guard, a script that forgets to set `opts.algorithm` after switching to a PEM key would silently sign an HS256 token using the PEM bytes themselves — that's exactly the algorithm-confusion attack pattern in reverse.
- **Plain bytes + asymmetric algorithm** throws with the inverse hint ("needs a PEM-encoded private/public key but secret is plain bytes"). Without this guard the user would get a cryptic "x509: malformed certificate" deep inside `jwt-go`.

The cross-check fires at the binding boundary before `jwt-go` is called, so it's a thrown error rather than a `valid: false` resolution — these are structural input problems, not validation failures.

- **`sign(claims, secret, opts?)`** — Produces a compact-serialised signed JWT. `claims` is passed straight through to `jwt-go`'s `MapClaims`, which recognises the RFC 7519 reserved claims (`exp`, `nbfi`, `iat`, `aud`, `iss`, `sub`, `jti`) and lets you layer arbitrary application claims alongside them. Missing reserved claims aren't synthesised — scripts that want `iat` set should compute it explicitly (e.g. `Math.floor(api.time.nowMs() / 1000)`).
- **`view(token)`** — Decodes the header + payload **without verifying the signature**. Useful for debugging auth flows or surfacing `aud` / `iss` to the user before deciding whether to trust the token. `signature` comes back as the raw base64url-encoded segment (no padding); decode it yourself if you need the bytes. Malformed input — wrong segment count, invalid base64, invalid JSON — throws.
- **`validate(token, secret, opts?)`** — Verifies the signature using `secret` and the algorithm declared in the token's header. Standard-claim validation (`exp`, `nbfi`, `iat`) is delegated to `jwt-go`. `opts.audience` / `opts.issuer` are cross-checked against `aud` / `iss` when set. The contract is **resolve, don't throw, on validation failure**: `{ valid: false, reason: "..." }` for bad signature, expired, audience mismatch, issuer mismatch, *and* algorithm mismatch when `opts.algorithm` is set. Only structural input errors (not a JWT at all, empty secret, empty token, PEM/algorithm cross-check failure) throw — pattern-matching on `valid: false` shouldn't accidentally accept a garbage string.

Setting `opts.algorithm` is the **algorithm-confusion guard**: an attacker who has the public key bytes can forge an HS256 token by using those bytes as the HMAC secret (the classic JWT exploit). When `opts.algorithm` is set, the parser's `WithValidMethods` whitelist refuses any token whose header declares a different algorithm. Always set

`opts.algorithm` when validating asymmetric tokens; the binding doesn't infer it from the key shape because that would make the wrong default silent.

Age encryption (`api.encrypt.*`) wraps filippo.io/age, the pure-Go reference implementation. The round-trip core (keygen / encrypt / decrypt) lands in v0.5.5; v0.5.6 adds ASCII-armoured output via `opts.armored`; v0.5.7 adds `rekey` for rotating recipients without exposing plaintext; v0.5.8 adds `detectBackend` for classifying recipient / identity strings. A future cut will extend `encrypt` / `decrypt` to also accept PGP recipients recognised by `detectBackend`. Identities use the X25519 flavour: `age1...` recipients (safe to share) and `AGE-SECRET-KEY-1...` identities (must be kept secret).

All three members are synchronous: encryption is pure CPU work with a small API surface, matching the call shape of the other crypto bindings (`api.jwt`, `api.hash`).

- **keygen()** — Generate a fresh X25519 identity. Returns `{ publicKey, privateKey }` as the bech32 strings age writes to disk. Two consecutive calls produce different keys; entropy comes from `crypto/rand` via age. Destructure to use one or the other separately.
- **encrypt(data, recipients, opts?)** — Seal `data` to one or more recipients. Input accepts `string` / `Uint8Array` / `ArrayBuffer`; `recipients` accepts a single string or an array. Default output is the binary age format as `Uint8Array`. Passing `opts.armored: true` wraps it in age's ASCII armor — the `-----BEGIN AGE ENCRYPTED FILE-----` banner + base64 body that's safe to embed in JSON / YAML / email. Either form decrypts via the same `decrypt` call (auto-detected by the leading bytes). Multi-recipient encryption uses age's native multi-recipient header — any one of the listed identities can decrypt the resulting ciphertext, no re-encryption per reader needed.
- **decrypt(ciphertext, identities)** — Open an age-encrypted payload with one of the supplied identities. **Auto-detects binary vs armored input** by sniffing the leading bytes for age's armor banner (whitespace and BOM bytes ahead of the banner are tolerated, so ciphertext pasted from JSON / email containers still detects correctly). age then walks the identities and uses the first that matches a stanza in the header. Returns `Uint8Array`; let scripts decode to a string via `api.text.decode(bytes, "utf-8")` if appropriate (goja doesn't ship `TextDecoder`).
- **rekey(ciphertext, oldIdentities, newRecipients, opts?)** — Re-encrypt a payload for a fresh recipient set without ever exposing the plaintext to the caller. The decrypt+encrypt loop is internal; the plaintext lives only in this function's stack until the encrypt step finishes. Output format defaults to **match the input** — armored in / armored out, binary in / binary out — which is what you almost always want when key-rotating a payload that lives in a fixed location (file, vault row, JSON field). `opts.armored: true` | `false` forces the output regardless of input. Internally consistent with `encrypt` and `decrypt`: same cross-checks (private as recipient, public as identity), same auto-detect on the input side, same multi-recipient handling on the output side.
- **detectBackend(input)** — Classify a recipient or identity string by encryption backend without parsing or I/O. Returns `{ backend: "age" | "pgp" | "unknown", kind?:`

"public" | "private" }. Age recognises three input forms: bech32 X25519 (age1..., classified kind: "public"), AGE-SECRET-KEY-1... (kind: "private"), and SSH public keys (ssh-rsa / ssh-ed25519, classified kind: "public" since age accepts them as recipients). PGP recognises armored block markers (-----BEGIN PGP PUBLIC KEY BLOCK----- / -----BEGIN PGP PRIVATE KEY BLOCK-----). Anything else returns { backend: "unknown" } — false-negative on unfamiliar input is the safer default than guessing. v0.5.8 ships the classifier only; sercon's actual encrypt / decrypt paths still handle age recipients only. Useful for scripts that need to dispatch on the format ("is this an age recipient or a PGP block?") before deciding which encrypt path — e.g., a config loader that routes some recipients through api.encrypt.encrypt and shells out to gpg --encrypt for others.

Cross-checks fire at the binding boundary so the common JS-side mistakes throw with named-key hints rather than cryptic bech32 errors deep inside age:

- **Private key as recipient** ("looks like a private key (AGE-SECRET-KEY-...); recipients are public keys (age1...)") — catches a script that mixed up the two halves on the encrypt call.
- **Public key as identity** ("looks like a public key (age1...); identities are private keys (AGE-SECRET-KEY-1...)") — inverse guard on the decrypt call.

Decryption with an identity that wasn't on the recipient list throws with age's own wording forwarded through: "identity did not match any of the recipients: incorrect identity for recipient block". Scripts that want to silently try multiple identities should pass them all in one call (age tries each in turn internally) rather than catching this error.

Hash bindings interpret the input as a UTF-8 byte sequence and return lowercase hex. SHA-3 functions are sha3_256 / sha3_512 (the IETF spec uses the underscore so the JS name matches recon's). BLAKE3 uses the upstream lukechampine.com/blake3 reference implementation with a 32-byte output. crc32 is the IEEE polynomial, zero-padded to 8 hex chars.

String utilities (api.str.*) follow PHP-/recon-style semantics where they differ from JS:

- trim / ltrim / rtrim accept an optional mask string; **any character in the mask** is stripped. Default mask is whitespace (" \t\n\r\v\f").
- reverse is rune-aware, so reverse("café") is "éfac". Grapheme clusters are not handled — that's a separate problem.
- nl2br emits
 by default; pass true as the second arg for XHTML-style
.
- urlEncode / urlDecode use the form encoding (+ for space). For path-segment encoding, use encodeURIComponent instead — that one is provided by goja natively.
- pad / lpad / rpad follow recon's str_pad: pad defaults to side "right"; "left" and "both" are also accepted.
- sprintf / printf are thin wrappers over Go's fmt, so the verb set is Go's (%s, %d, %x, %.2f, %v, %t, %q, etc.) — not PHP's. printf writes to stdout.

- `normalizeNewlines` canonicalises any mix of `\r\n`, `\r`, and `\n` to the requested style.

Path utilities (`api.path.*`) are POSIX (forward-slash). On Windows either pass forward-slash paths or convert separators yourself.

Time formatting (`api.time.format(unixMs, fmt, tz?)`) takes Unix milliseconds (symmetric with `api.time.nowMs()`) and a small strftime token set: `%Y %y %m %d %H %M %S %F %T %j %A %a %B %b %z %Z %%`. Day and month names are in English. Pass a third-argument IANA zone name (e.g. `"UTC"`, `"America/New_York"`) to render in that zone; if omitted, the host's `time.Local` is used. Unknown `%X` tokens are emitted verbatim so a typo is visible rather than silently dropped.

Protocol probes (`api.net.*`) are stdlib-backed and hit the real network. All three return Promises (via `PromisifyAsync`) and accept an optional `{ timeout: <ms> }` second arg; the default is 5 seconds.

- **`tcp(target, opts?)`** — dials TCP, measures round-trip from resolution to handshake. `target` is `"host:port"` or just `"host"` (with `opts.port` overriding the default `80`). Resolves to the remote IP, port, and `latencyMs`.
- **`dns(host, opts?)`** — looks up A, AAAA, MX, TXT, CNAME, and NS records via `net.Resolver`. Pass `opts.types` to scope the query; empty record sets are omitted from the result object so scripts can probe membership with `if ("mx" in result)`.
- **`tls(target, opts?)`** — connects with `InsecureSkipVerify` so even expired or hostname-mismatched certs come back inspectable. Returns the leaf cert's CN, issuer CN, validity window, `daysRemaining`, `dnsNames`, serial number, and a SHA-256 fingerprint (lowercase hex). Default port is `443`. Hosts that care about validity should re-validate the cert themselves.
- **`ntp(host, opts?)`** — queries an NTPv4 server (UDP 123 by default) via github.com/beevik/ntp. Returns the server time (ISO 8601), local-clock offset, round-trip time, stratum, reference time, and the root delay / dispersion values from the server's response. All durations are in milliseconds with sub-millisecond precision.
- **`whois(domain, opts?)`** — two-hop WHOIS lookup (IANA → registrar's whois server) via github.com/likexian/whois with parsing through github.com/likexian/whois-parser. `raw` is always populated; `domain` and `registrar` are best-effort parsed views (some TLDs the parser doesn't understand will return only `raw`). The whois library doesn't accept a `context.Context` — `opts.timeout` is plumbed through its own per-client setting and the engine's `Options.Timeout` watchdog catches the outer call.

Email-authentication probes (`api.email.*`) read DNS records and surface the published policy. They all return `{ present: false }` when the relevant record is absent (NXDOMAIN or no TXT record matching the marker prefix), so scripts can write a single presence-check pattern across the family.

- **`spf(domain)`** — queries `TXT(<domain>)` and returns the first record starting with `v=spf1`. `mechanisms` is the tokenised sequence after the version prefix; `allPolicy`

summarises the trailing `all` qualifier ("pass" for `all` / `+all`, "fail" for `-all`, "softfail" for `~all`, "neutral" for `?all`, empty string when no `all` is present).

- **`dmARC(domain)`** — queries `TXT(_dmarc.<domain>)` and parses the `v=DMARC1; key=val; ...` form into a flat tag map (keys are case-folded; values retain internal whitespace and commas). `policy` / `subdomain` / `percent` / `rua` / `ruf` are surfaced separately because those are the fields most scripts care about.
- **`mtaSts(domain)`** — looks up `TXT(_mta-sts.<domain>)` and, on success, fetches the policy file at `https://mta-sts.<domain>/.well-known/mta-sts.txt`. The TXT carries a versioned `id` for change detection; the policy file is where the actual `mode` + `mx` list live. RFC 8461 format (line-based, `key: value`, `mx: repeats`) is parsed inline. Policy-fetch failures (TLS errors, 4xx, timeout) don't fail the binding — the TXT part is still returned, and the fetch error surfaces under `policyError` so scripts can decide what to do.
- **`tlsRpt(domain)`** — looks up `TXT(_smtp._tls.<domain>)` and parses the `v=TLSRPTv1; rua=...` form into a tag map. `rua` is surfaced separately because that's the actionable bit.
- **`bimi(domain, opts?)`** — looks up `<selector>._bimi.<domain>`, selector defaulting to `default`. Surfaces the logo URL (`l`) and VMC URL (`a`) from the tag map.
- **`all(domain)`** — runs every probe above in parallel via goroutines and returns a single aggregate object keyed by probe name. Per-probe failures don't fail the aggregate — they surface under `<probe>.error` so a partial result is still useful (e.g. SPF + DMARC found, MTA-STS policy fetch timed out).

6. JavaScript runtime built-ins (goja)

`scriptengine` runs on goja, which implements **ES5.1 + a large subset of ES6+**. The following are present and behave per spec:

Globals

- `Object`, `Array`, `String`, `Number`, `Boolean`, `BigInt`
- `Math`, `Date`, `JSON`, `RegExp`
- `Error`, `TypeError`, `RangeError`, `SyntaxError`, `ReferenceError`, `EvalError`, `URIError`
- `Promise` (provided via goja's native implementation; resolution microtasks are pumped by the event loop)
- `Symbol` (partial: well-known symbols `Symbol.iterator`, `Symbol.asyncIterator`, etc., are supported)
- `Proxy`, `Reflect` (partial)

```
api.log(Math.PI.toFixed(4), Math.max(1, 9, 3));
api.log(new Date().toISOString());
api.log(JSON.stringify({ a: 1, b: [2, 3] }));
api.log("abc".repeat(3), "abc".padStart(6, "_"));
```

Collections

- `Map`, `Set`, `WeakMap`, `WeakSet`

```
const counts = new Map<string, number>();
for (const ch of "banana") counts.set(ch, (counts.get(ch) ?? 0) + 1);
api.log([...counts]); // [{"b",1}, {"a",3}, {"n",2}]
```

Typed arrays

- `ArrayBuffer`, `DataView`
- `Int8Array`, `Uint8Array`, `Uint8ClampedArray`, `Int16Array`, `Uint16Array`, `Int32Array`, `Uint32Array`, `Float32Array`, `Float64Array`

```
const buf = new ArrayBuffer(4);
new DataView(buf).setUint32(0, 0xCAFEBAFE);
api.log(new Uint8Array(buf)[0].toString(16)); // ca
```

Iteration and generators

- `for...of`, `for...in`, `spread`, `destructuring`
- Generator functions (`function*`)
- Async generators (`async function*`) — supported via goja's event loop

```
function* fib() {
  let [a, b] = [0, 1];
  while (true) {
    yield a;
    [a, b] = [b, a + b];
  }
}
const it = fib();
const first10 = Array.from({ length: 10 }, () => it.next().value);
api.log(first10);
```

Not (yet) provided

- `fetch` — use `api.http.*` from the CLI, or register your own binding.

- `Worker`, threading primitives.
- DOM globals (`window`, `document`).
- Native ES modules (`import` / `export` are *transpiled* to CommonJS at load time — see [section 8](#)).

7. Async runtime additions (goja_nodejs)

The event loop bundles a small Node-compatible runtime on top of goja:

Timers

```
setTimeout(() => api.log("after 50ms"), 50);
setInterval(() => api.log("tick"), 100);
setImmediate(() => api.log("next tick"));
clearTimeout(t); clearInterval(i); clearImmediate(im);
```

The event loop holds the script alive while *any* timer is pending. If your script ends with only `setTimeout` callbacks scheduled, the run will wait for them to fire before returning.

console

```
console.log("info");
console.error("oops");
console.warn("careful");
console.info("fyi");
console.debug("trace");
```

Disable with `Options.DisableConsole = true` if you want a clean global namespace.

require

CommonJS `require`, with TS-aware extension fallback added by `sercon` (see [section 10](#)):

```
const helpers = require("./helpers");
helpers.doThing();
```

Both `require("./foo")` and `import { x } from "./foo"` resolve to the same module instance per Run; esbuild rewrites `import` to `require` at transpile time.

8. TypeScript support

TS is transpiled by esbuild in-process. There is no `tsc`, no `tsconfig.json`, no type-checking — types are erased and the resulting JS is what executes. Practical implications:

- All TS syntax esbuild supports is allowed: type annotations, `interface`, `type`, generics, enums (compiled to objects), namespaces, decorators (legacy & TC39 stage 3).
- `import type { X } from "y"` is stripped entirely (no runtime require).
- TS errors are *not* surfaced as build errors — only esbuild parse errors are. Run a separate `tsc --noEmit` in CI if you want type enforcement.
- `tsconfig.json` is **not** consulted.

`.tsx` and `.jsx` are supported via esbuild's `LoaderTSX` / `LoaderJSX`, including for required modules resolved by the engine's extension fallback. JSX has no React runtime in scope by default — either set the factory at the file level with an `@jsx` pragma:

```
/** @jsx h */
function h(tag: string, props: any, ...children: any[]) {
  return { tag, props: props ?? {}, children };
}
export const Box = (label: string) => <div className="box">{label}</div>;
```

...or provide a `React.createElement`-compatible binding from Go and rely on esbuild's default pragma. ESM `export default <value>` also works through the entry-script rewriter's `__esModule ? .default : m` unwrap, so `import answer from "./mod"` resolves to the default export.

9. Top-level `await`

Top-level `await` works in entry scripts:

```
const r = await api.http.get("https://example.com");
api.log(r.status);
```

How it works under the hood: esbuild rejects top-level `await` with the `cjs` output format, so the engine transpiles the *entry* script with `format=esm`, then rewrites `import` statements to `require()` calls and wraps the rest of the body in:

```
;(async () => { /* your transpiled body */ })().then(__resolve, __reject);
```

`__resolve` and `__reject` are bindings the engine sets on the runtime to capture the final settlement. *Required-module* code (anything loaded through `require`) is transpiled with `format=cjs` and does **not** support top-level `await` — wrap in `(async () => ...)()` yourself if you need it inside a module.

10. Module resolution

The engine plugs a custom source loader into `goja_nodejs/require` that adds:

1. **Direct path**: if the requested file exists on disk, use it. `.ts` / `.tsx` files are transpiled on the fly.
2. **Extension fallback** when the request has no extension: try `.ts`, `.tsx`, `.js`, `.cjs`, `.mjs`, `.json` in that order.
3. **.js → .ts swap**: if a path ending in `.js` / `.cjs` / `.mjs` is requested but the literal file doesn't exist, the same path ending in `.ts` (or `.tsx`) is tried. Handles `package.json` `main` fields that point at compiled output where only the TypeScript source is on disk.
4. **package.json source preference**: when reading a `package.json`, if a `source` field is present and points at an existing `.ts` / `.tsx` file, `main` is rewritten to that path before the registry consumes it. Matches the convention used by parcel, microbundle, and similar bundlers to mark the TS source of truth.
5. **Directory index**: if the resolved path is a directory, try `index.ts`, `index.tsx`, `index.js`.

The base directory follows Node's CommonJS rules — relative imports resolve against the requiring module's directory, courtesy of `goja_nodejs`.

```
// All resolve to ./helpers/assert.ts:
import { check } from "./helpers/assert";
import { check } from "./helpers/assert.ts";
const { check } = require("./helpers/assert");
const { check } = require("./helpers/assert.js");
```

JSON imports work out of the box via either the ESM-style default import (esbuild rewrites it to a `require`) or a direct `require`:

```
import data from "./data.json";
const r = require("./data.json");
```

`node_modules` lookup *works* via the upstream registry, but the source loader doesn't currently transpile through that path — keep TS helpers under `ScriptRoot`.

11. Timeouts and cancellation

Two mechanisms interrupt a running script:

- `Options.Timeout` — wall-clock limit per `Run`. Exceeding it returns `scriptengine.ErrScriptTimeout`.

- `ctx` passed to `Run / RunFile` — cancelling the context returns `ctx.Err()` (typically `context.Canceled` or `context.DeadlineExceeded`).

Both call `vm.Interrupt(...)` and `loop.Terminate()` from a watcher goroutine. This works for **sync** JS — including `while(true){}` — because the goja VM checks the interrupt flag between bytecode steps. A host Go binding that's blocked in cgo or a long syscall isn't interruptible mid-call; if you write such a binding, plumb the engine `ctx` through to it yourself.

```
eng := scriptengine.New(scriptengine.Options{Timeout: 200 *
time.Millisecond})
_, err := eng.Run(ctx, "loop.ts", `while (true) {}`)
// err is scriptengine.ErrScriptTimeout, returned ~200–300ms after Run.
```

12. Error semantics

- A Go binding returning `(T, error)` automatically throws as a JS exception when `err != nil`. The exception's `.message` is `err.Error()`.

```
try { mayFail(); } catch (e) { api.log(String(e)); }
```

- A Go binding that `panic`s with a `*goja.Object` (e.g. `vm.NewGoError(...)`) does the same.
- A script throwing a JS error out of the top level surfaces from `Run` as a Go `error` whose message is the JS error's `.message`.
- Timeout errors satisfy `errors.Is(err, scriptengine.ErrScriptTimeout)`. Cancellation errors satisfy `errors.Is(err, context.Canceled)` or `context.DeadlineExceeded`.

13. Type generation (.d.ts)

`Engine.WriteTypes(w)` walks the registered bindings and emits a TypeScript declaration file. The mapping table (abbreviated):

Go type	TS type
<code>string</code>	<code>string</code>
<code>any numeric / float64</code>	<code>number</code>
<code>bool</code>	<code>boolean</code>
<code>[]T</code>	<code>T[]; []byte → Uint8Array</code>
<code>map[string]T</code>	<code>Record<string, T></code>

Go type	TS type
<code>*T</code>	<code>T</code> (transparent unwrap)
<code>error</code> (single return)	omitted (collapses to <code>void</code>)
<code>(T, error)</code> (tuple return)	<code>T</code>
<code>interface{}</code> / <code>any</code>	<code>unknown</code>
<code>goja.FunctionCall</code> parameter	folded into <code>(...args: unknown[])</code>
<code>goja.Value</code> return	<code>unknown</code>
<code>scriptengine.AsyncBinding</code> value (from <code>PromisifyAsync[T]</code>)	<code>Promise<T></code>
self-referential types	<code>unknown</code> (cycle detection bails after depth 4)

```
sercon --emit-dts api.d.ts
```

In your editor, place the resulting `.d.ts` next to your scripts (or reference it via `/// <reference path="..." />`) for autocomplete.

JSDoc comments on declarations

The emitter pulls JSDoc strings from the engine's doc map (set via `Engine.SetDocs` / `Engine.SetMemberDocs`) and renders them above each declaration. Single-line docs collapse to `/** ... */`; multi-line docs expand to a standard `/*`-prefixed block. Bindings without a doc entry emit no JSDoc — the emitter never inserts empty placeholders. Docs are looked up by the same dotted path the binding was registered under, so namespace members get hover text too:

```
// AUTO-GENERATED by scriptengine.Engine.WriteTypes – do not edit by hand.

/** sercon's built-in script surface. ... */
declare const api: {
  hash: {
    /** SHA-256 hex digest of a UTF-8 input. */
    sha256(...args: unknown[]): string;
    /** BLAKE3 hex digest (32-byte output, lukechampine.com/blake3). */
    blake3(...args: unknown[]): string;
    // ...
  };
  // ...
};
```

The CLI's `api.*` surface ships with a curated doc map; see `cmd/sercon/api_docs.go` for the source of truth and add new entries there when adding new bindings (the lockstep rule keeps `--examples / MANUAL / api.d.ts` in sync, and the doc map is now the seventh artifact in that chain).

14. Limitations and gotchas

- **No native ES modules at runtime.** All `import` is rewritten to `require` by esbuild. Side-effect-only imports run the module body exactly once per Run.
- **No top-level await inside required modules.** Wrap with `(async () => ... )()` inside the module if you need it.
- **No `tsconfig.json` honoured.** Module resolution and type checking are independent from any project-level TS config.
- **Per-Run runtime.** Side effects on `globalThis` do not persist between calls to `Run` on the same `Engine`. The require *compile* cache is shared; module *exports* are not.
- **Multi-line ESM imports may stress the entry rewriter.** The current rewriter is line-based and handles the common forms; especially gnarly inputs (comments inside the spec list, very unusual whitespace) fall back to executing as-is, which may throw.
- **RegisterConstructor is d.ts-only today.** At runtime it behaves like `Register`. True `new`-able constructor semantics are on the roadmap.
- **HTTP bindings are real network calls.** `api.http.*` uses `net/http` with a 5s timeout. They are not mockable from JS.

See [OUT-OF-SCOPE.md](#) for the active backlog of deferred ideas.

This manual covers sercon v0.5.8. Whenever you add, remove, or change a flag, a binding, or the script API, update this file alongside the help screen (`--help`), the examples walkthrough (`--examples`), and the `CHANGELOG.md`.